SCoPE-MS: New Technique for Quantifying Proteomes of Single Mammalian Cells

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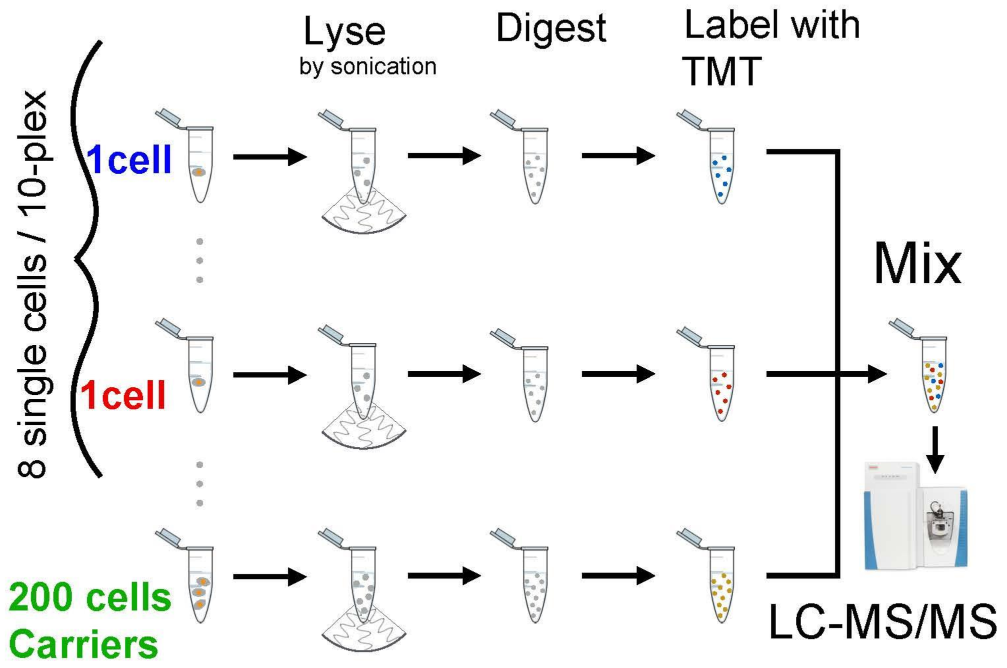
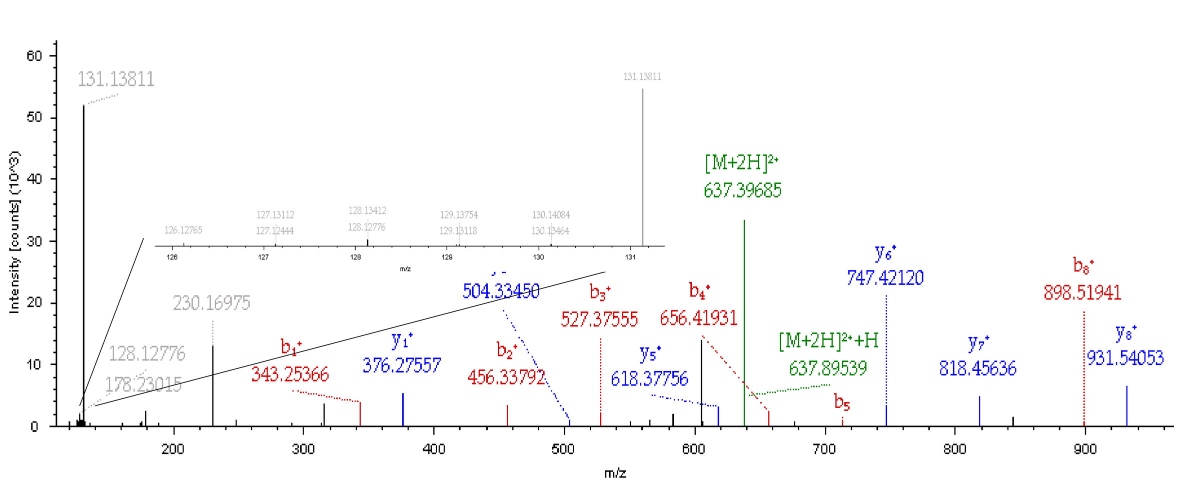
Typical work flow of SCoPE-MS. Individually picked live cells are lysed by sonication, the proteins in the lysates are digested with trypsin, the resulting peptides labeled with TMT labels, combined and analyzed by LC-MS/MS in Orbitrap instrument.

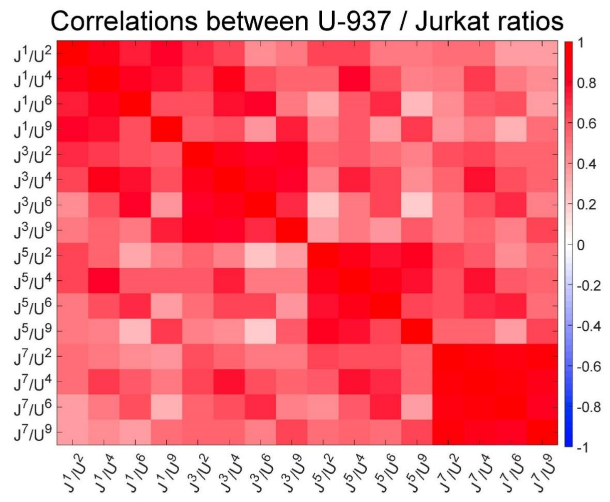


Experimental Design Table

Label	Set 1	Set 2	Set 3
126	U-937	Jurkat	Jurkat
127N	Jurkat	U-937	Jurkat
127C	U-937	Jurkat	Jurkat
128N	Jurkat	U-937	Jurkat
128C	U-937	Jurkat	U-937
129N	Jurkat	U-937	U-937
129C	U-937	Jurkat	U-937
130N	*empty*	*empty*	*empty*
130C	Jurkat	U-937	U-937
131 Carriers	100 Jurkat 100 U-937	100 Jurkat 100 U-937	100 Jurkat 100 U-937

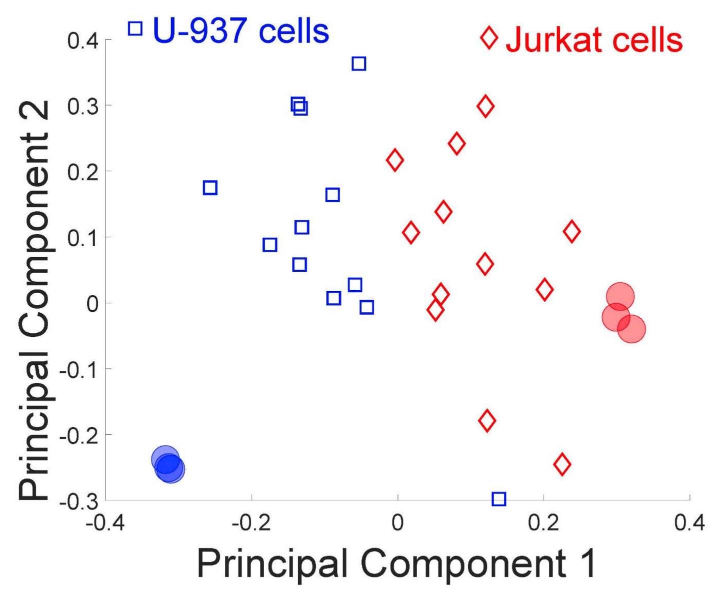
Typical MS/MS spectra of TMT labeled single cells set with carrier channel and correlation between Jurkat and U927 cell labeled with TMT in corresponding channels.

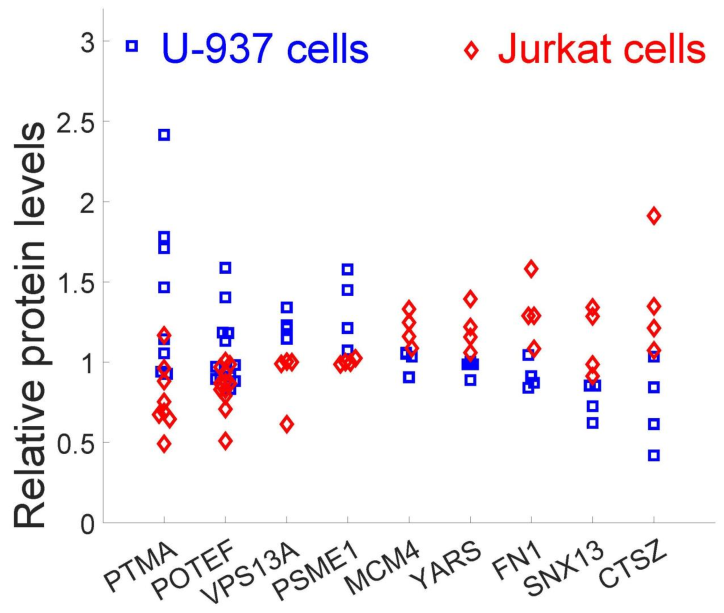




Unsupervised principal component (PC) analysis using data for quantified proteins from the experiments described above stratifies the proteomes of single cancer cells by cell type. Protein levels from 6 bulk samples from Jurkat and U-937 cells are also projected and marked with filled semitransparent circles.

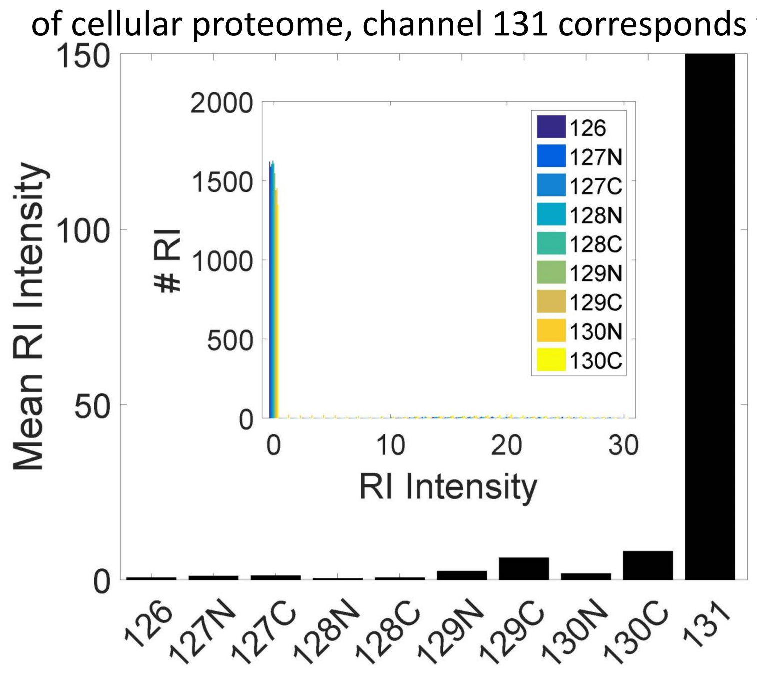
Distributions of protein levels across single U-937 and Jurkat cells indicate cell-type-specific protein abundances.

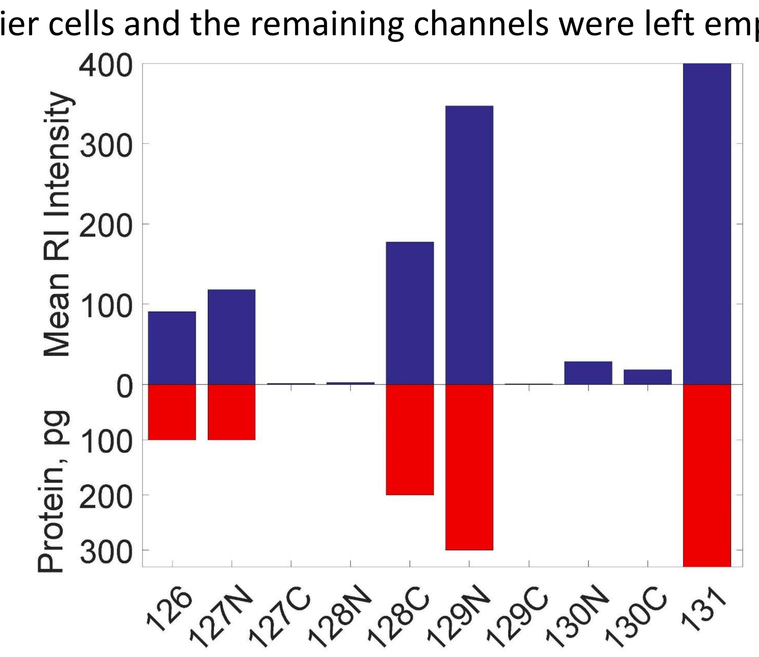




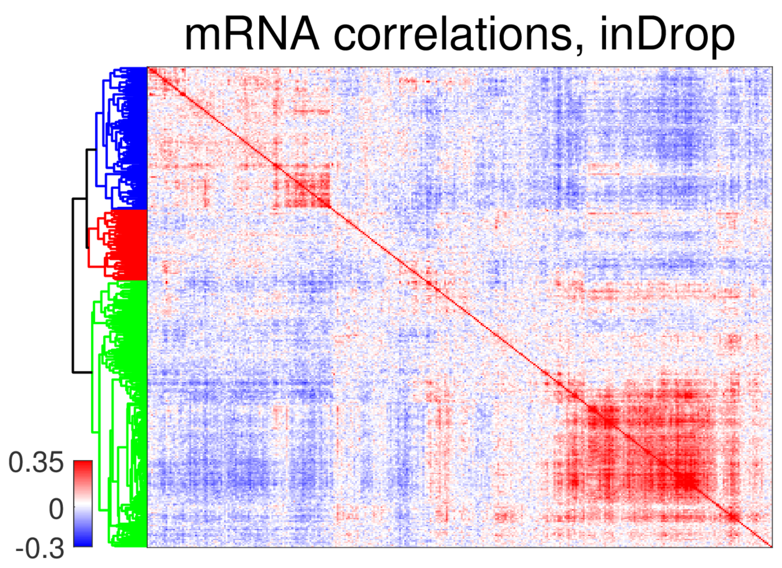
Contribution of background noise to quantification of peptides in single cells. Reporter ion (RI) intensities in a SCoPE set in which the single cells were omitted while all other steps were carried out, i.e., trypsin digestion, TMT labeling and addition of carrier cells in channel 131.

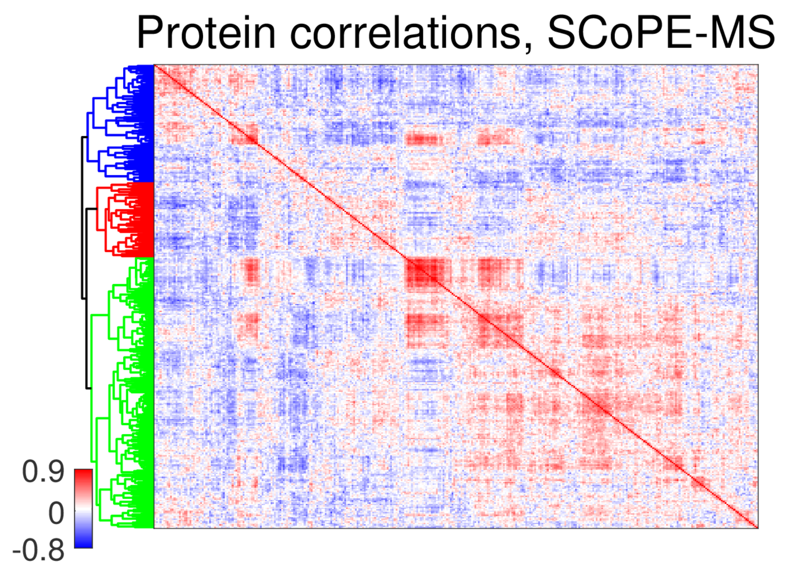
Mean RI intensities for a TMT set in which only 6 channels contained labeled proteome digests and the other 4 were left empty. Channels 126, 127N, 128C, and 129N correspond to peptides diluted to levels corresponding to 100, 100, 200 and 300 picograms of cellular proteome, channel 131 corresponds to the carrier cells and the remaining channels were left empty.





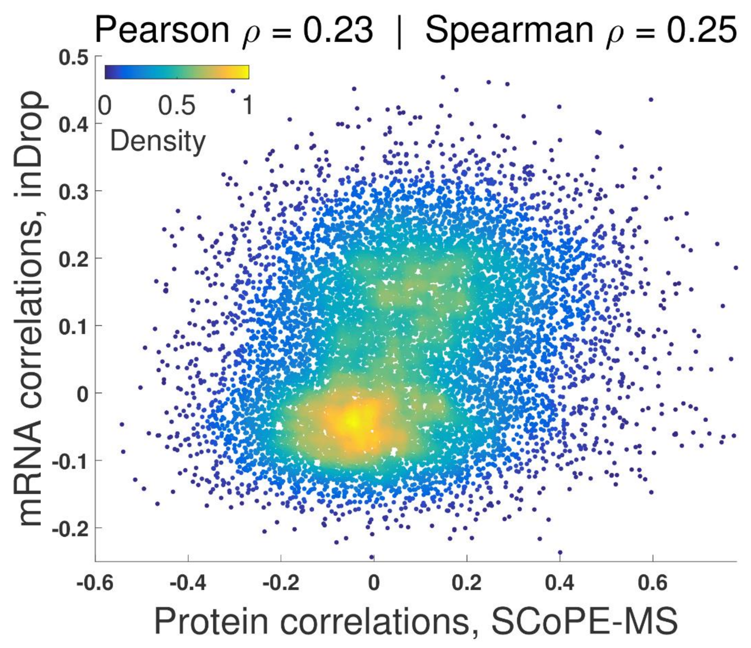
Clustergram of pairwise correlations between mRNAs with 2.5 or more reads per cell as quantified by inDrop in single EB cells and pairwise correlations between proteins quantified by SCoPE-MS in 12 or more single EB cells. The overlap between corresponding RNA from inDrop and protein clusters from SCoPE-MS indicates similar clustering patterns.

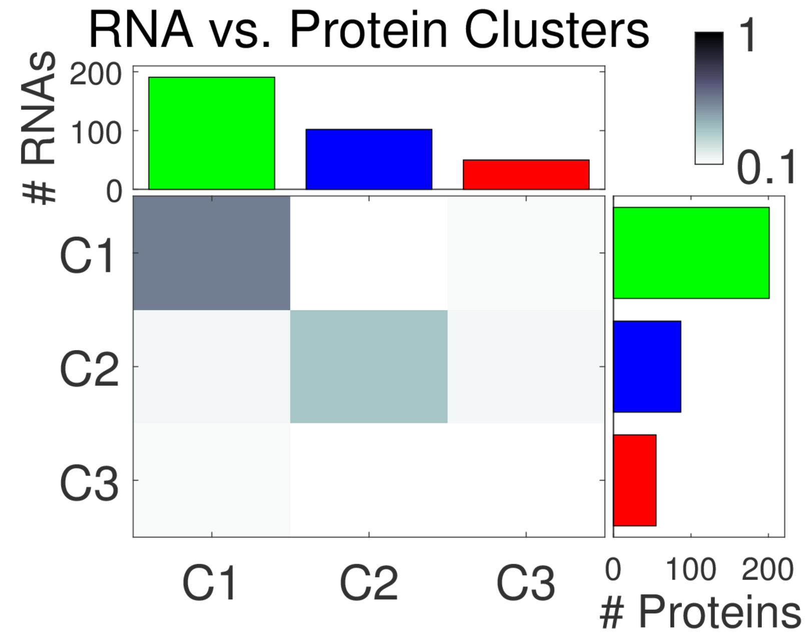




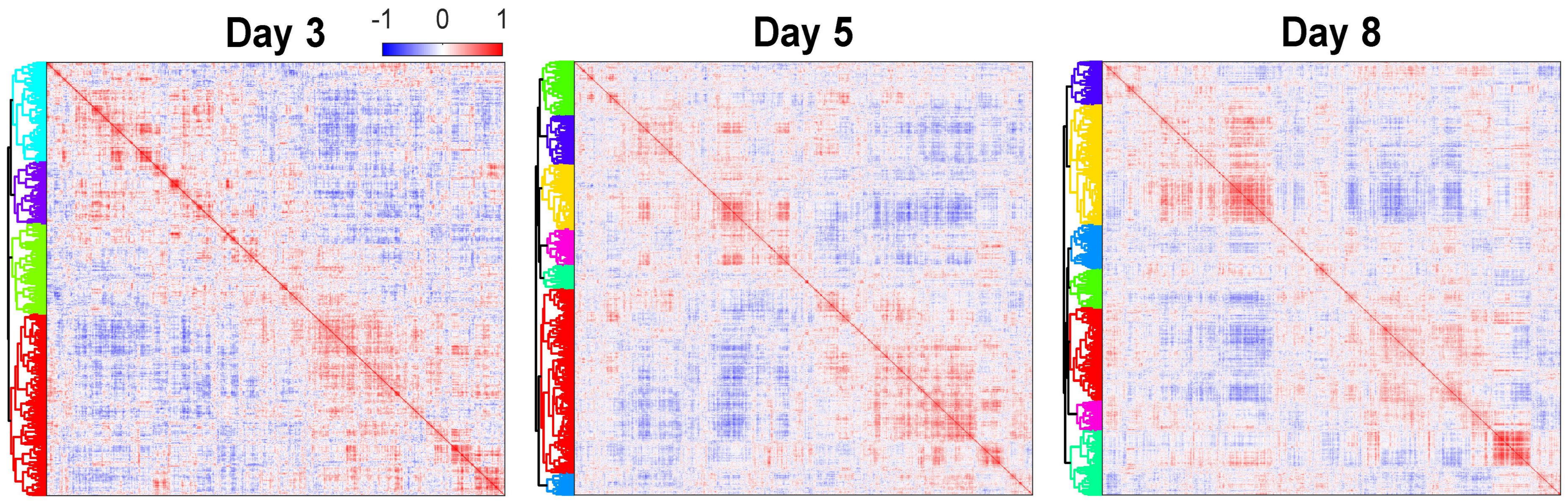
Protein-protein correlations correlate to their corresponding mRNA-mRNA correlations. Only genes with significant mRNA-mRNA correlations were used for this analysis.

The overlap between corresponding RNA from (a) and protein clusters from (b) indicates similar clustering patterns.



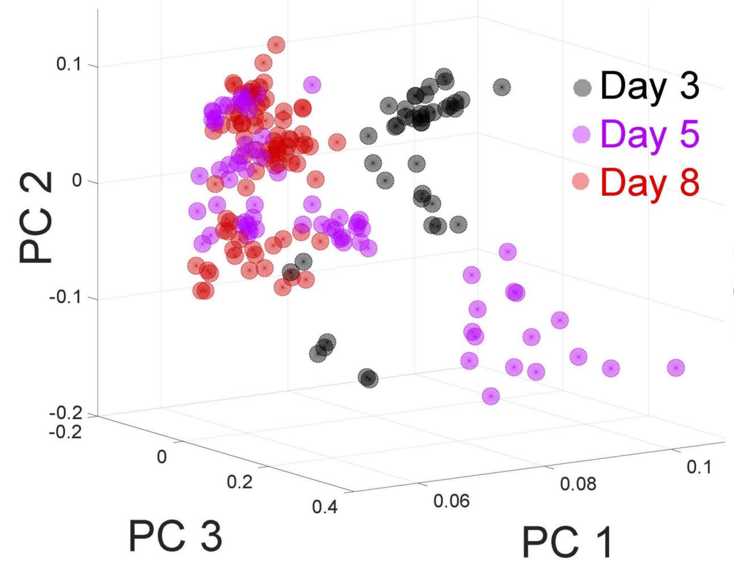


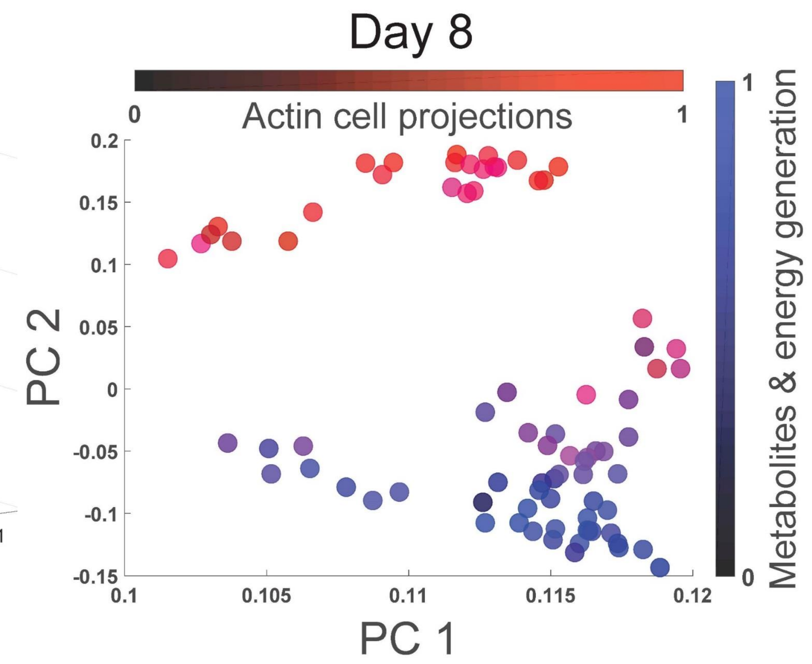
Clustergrams of pairwise protein-protein correlations in cells differentiating for 3, 5, and 8 days after LIF withdrawal. The correlation vectors were hierarchically clustered based on the cosine of the angles between them.

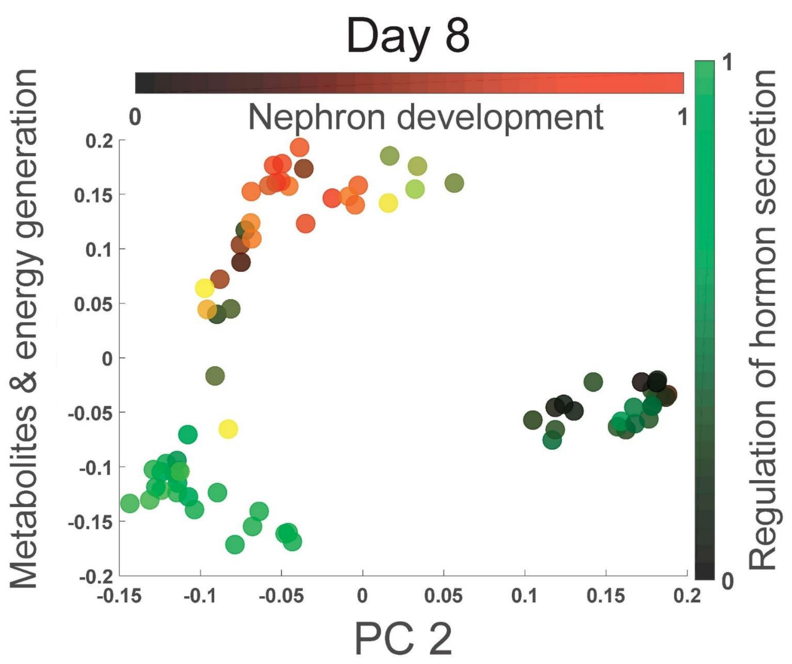


The proteomes of all single EB cells were projected onto their PCs, and the marker of each cell color-coded by day. The single-cell proteomes cluster by day, a trend also reflected in the distributions of PC 1 loadings by day.

The proteomes of cells differentiating for 8 days were projected onto their PCs, and the marker of each cell color-coded based on the normalized levels of all proteins from the indicated gene-ontology groups.

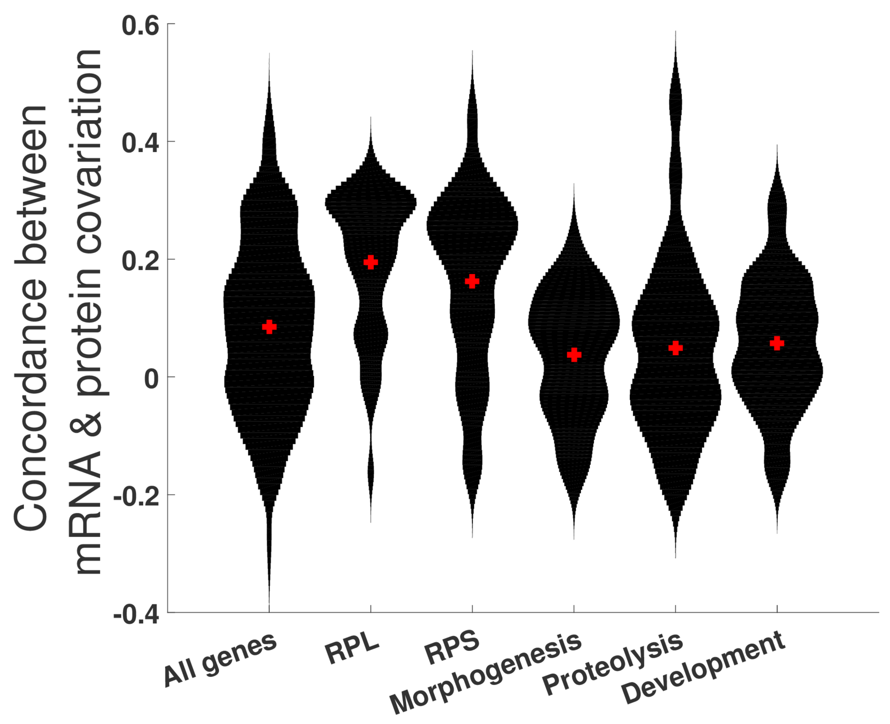


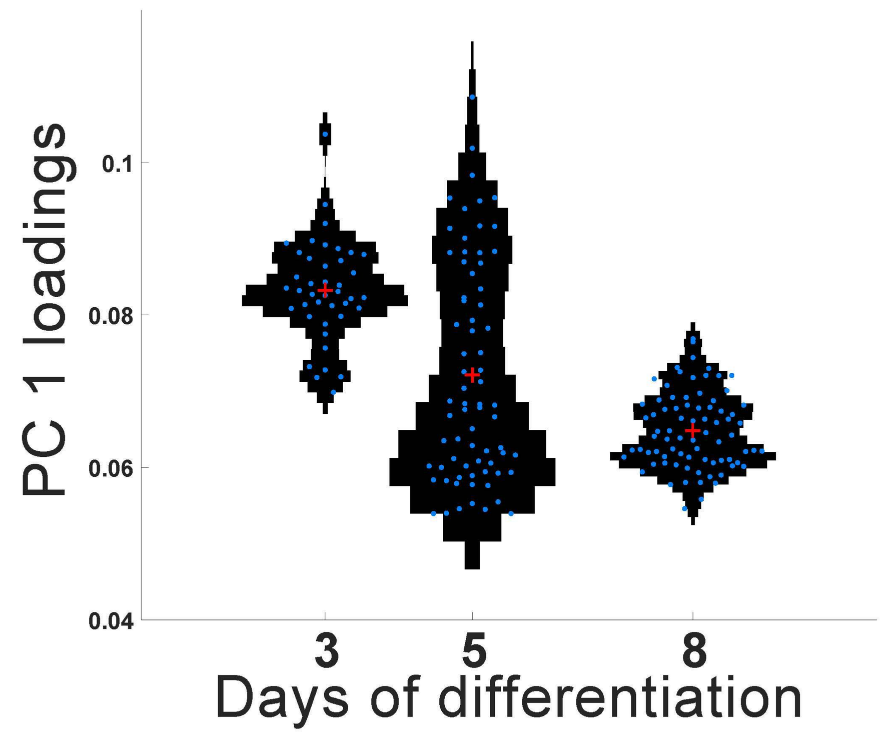




The proteomes of all differentiating single cells were decomposed into singular vectors and values, and distributions of the loading (elements) of the singular vector with the largest singular value, i.e., PC 1, shown as violin plots. Individual blue circles correspond to single cells, and the red crosses correspond to the medians for each day.

The concordance between corresponding mRNA and protein correlations (computed as the correlation between corresponding correlations) is high for ribosomal proteins (RPL and RPS) and lower for developmental genes; distribution medians are marked with red pluses. Only the subset of genes quantified at both RNA and protein levels were used for all panels.





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